

An M -augmenting path is a path

$$\begin{array}{ccccccc} v_1, v_2, v_3, v_4, \dots, v_{2K} \\ \text{┌} & & & & & & \\ (v_1, v_2) & (v_3, v_4) & \dots & (v_{2k-1}, v_{2k}) \\ | & | & & | \\ \mathcal{E}_M & \mathcal{E}_M & & \mathcal{E}_M \end{array}$$

Maximum Matching Characterization

Thm: M is a maximum size matching
iff $\nexists$ an M -aug. path.

pf: " $\Rightarrow$ " obvious. // Exercise.

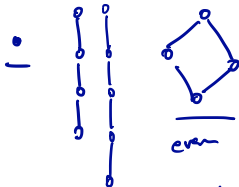
" $\Leftarrow$ " $\left[\begin{array}{l} \text{If } M \text{ not maximum size, then we can} \\ \text{find an } M\text{-aug. path.} \end{array} \right]$

M not max size, then $\exists M'$ matching, with $|M'| > |M|$.

$$\hat{G} = (V, M \cup M')$$

Deg. of each vertex in $\hat{G}$ is at most 2.

$\Rightarrow \hat{G}$ is a union of:



because $|M'| > |M|$

$\exists$ path of odd # edges
that starts & ends with
edge of M'

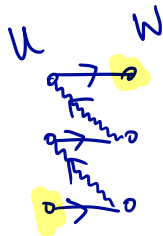
$\Rightarrow$ This is an augmenting path.

Application to Bipartite Matching

$G = (V, E)$ bipartite: $V = U \cup W$

Given matching M orient edges in E : $e = (u, w)$

orient $e = (u, w)$: from u to w
if $e \in M$



from w to u if $e \in M$.

Let $u_M \in U$, $w_M \in W$ be nodes in
 U, W resp'ly, not covered by M .

$\exists$ M -aug path iff $\exists$ directed path from u_M to w_M .
(Exercise).

If it exists, this can be found efficiently.

Max Matching - Bipartite

Thm: $G = (V, E)$ bipartite

$$V = U \cup W$$

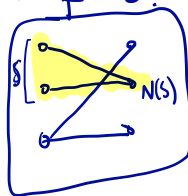
Then G has a matching covering U

iff $|N(S)| \geq |S| \quad \forall S \subseteq U$

if $S \subseteq U$, $N(S) \subseteq W$, is all neighboring nodes of S .

pf: " $\Rightarrow$ " obvious.

" $\Leftarrow$ " König's Thm says $\forall$ bipartite graph



size of max matching = size of min vertex cover.

Thus: enough to show that each vertex cover C has
 $|C| \geq |U|$

Now :



$$N(u \setminus C) \subseteq C \cap W$$

$$|C| = |C \cap U| + |C \cap W|$$

$$\geq |C \cap U| + |N(u \setminus C)|$$

$$\geq |C \cap U| + |u \setminus C| \quad \text{— by assumption}$$

$$= |u|.$$

